Supplementary Information

Sparse autoencoders reveal organized biological knowledge but minimal regulatory logic in single-cell foundation models

Ihor Kendiukhov

Supplementary Tables

Table S1: **Composition of the control cohorts.** Cells per line.

Cohort	K562	RPE1	Jurkat	HepG2	Total
Non-targeting cells in the resource	10,691	11,485	12,013	4,976	39,165
Atlas extraction cohort	581	567	591	261	2,000

The control condition is defined by perturbation label, so the cohort spans every cell line in the resource. The cohort reported here was reproduced from the extraction procedure under its recorded seed and validated against the per-cell gene counts recorded when the activations were written.

Table S2: **Per-ontology enrichment counts across all 18 Geneformer layers.** Each entry is the number of significant enrichments (FDR < 0.05) for features at that layer. GO BP = Gene Ontology Biological Process. TRRUST columns show enrichment for TF target sets (TF) and individual TF→target edges (Edges).

Layer	GO BP	KEGG	Reactome	STRING	TRRUST TF	TRRUST Edges
0	10,153	2,650	11,001	302	155	42
1	10,022	2,433	10,512	258	164	48
2	9,948	2,495	10,790	283	150	32
3	9,726	2,514	9,525	273	157	37
4	8,537	2,045	9,195	248	133	30
5	7,695	1,845	8,189	216	136	24
6	7,180	1,555	7,871	182	110	28
7	6,628	1,637	7,080	181	125	30
8	6,850	1,570	7,169	199	90	27
9	7,299	1,643	7,880	207	103	29
10	8,461	1,751	9,247	214	128	30
11	8,785	2,089	8,957	227	112	31
12	8,217	1,915	8,856	210	117	35
13	7,158	1,686	8,393	202	101	28
14	7,615	1,595	8,412	221	97	27
15	6,790	1,520	7,135	150	126	34
16	7,040	1,781	7,172	158	87	25
17	7,002	1,762	6,869	193	131	25
Total	**145,106**	**34,486**	**154,253**	**3,924**	**2,222**	**562**

Table S3: **Cross-layer feature persistence from layer 0.** Matches = features at L0 with cosine similarity > 0.7 to any feature at the target layer. The model undergoes radical representational transformation: by layer 6, essentially all features are novel with no L0 ancestry.

L0 → Target	Matches (cos > 0.7)	Rate
L0 → L1	114	2.5%
L0 → L2	93	2.0%
L0 → L4	67	1.5%
L0 → L6	25	0.5%
L0 → L8	10	0.2%
L0 → L10	1	∼0%
L0 → L12+	0	0%

Table S4: **Co-activation module statistics across all 18 layers.** PMI-based graphs with Leiden clustering (resolution = 1.0). Modules = number of distinct communities. Coverage = fraction of alive features in at least one module.

Layer	Modules	Feats in Modules	PMI Edges	Coverage
0	6	4,577	446,324	99.3%
1	8	4,562	440,681	99.0%
2	7	4,536	404,403	98.6%
3	8	4,518	393,574	98.3%
4	9	4,502	393,194	98.2%
5	12	4,472	390,845	97.7%
6	7	4,458	383,033	97.3%
7	8	4,439	371,832	96.7%
8	7	4,478	369,280	97.6%
9	7	4,535	380,304	98.7%
10	9	4,571	388,498	99.3%
11	8	4,565	388,103	99.3%
12	7	4,567	388,977	99.5%
13	7	4,561	383,779	99.5%
14	8	4,543	379,595	99.5%
15	8	4,461	340,269	98.2%
16	8	4,358	327,895	96.0%
17	7	4,474	343,059	97.7%
Total	**141**			

Table S5: **Geneformer cross-layer information highways.** PMI between SAE feature activations at source and target layers (500K positions each). A highway = source feature with $\geq$1 target-layer feature at PMI > 3. **Note:** the numbers here use the main-atlas 500K-position protocol; Table S6 reports the same three pairs recomputed on a 100K-position subsample with a permutation-null control. The two sets of numbers differ by less than one percentage point (e.g. L0$\rightarrow$L5: 98.4% here vs. 98.96% in Table S6) because of the different subsample sizes, and the null correction does not change the 100K numbers.

Layer Pair	Feats w/ Deps	Mean Max PMI	Median Max PMI	Max PMI	Highways
L0 $\rightarrow$ L5	4,604	6.61	6.72	11.10	4,530 (98.4%)
L5 $\rightarrow$ L11	4,518	6.63	6.71	10.87	4,401 (97.4%)
L11 $\rightarrow$ L17	4,555	6.79	6.86	10.66	4,544 (99.8%)

Table S6: **Null-corrected cross-layer highway fractions for the three long-range Geneformer pairs.** 100K gene positions subsampled from the K562 activation cache, encoded through the per-layer SAEs, cross-layer PMI computed as in Methods. Columns: observed 99th percentile of cross-layer PMI, permutation-null 99th percentile (τ_{null}^{p99}, max across 20 column-shuffle rounds), effective threshold $\tau = \max(3.0, \tau_{\text{null}}^{p99})$, and highway fraction. The permutation null's 99th-percentile PMI is *below* 3.0 in every pair tested, so the fixed threshold τ=3.0 is already conservative and the null correction leaves the highway fraction unchanged. This argues that the ∼97–99% highway rate is not an artifact of the k=32 TopK sparsity.

Src	Tgt	Obs $p99$ PMI	Null $p99$ PMI	τ_{eff}	Fixed-τ highway%	Null-corr highway%
L0	L5	4.76	2.46	3.00	98.96%	98.96%
L5	L11	4.93	2.76	3.00	96.66%	96.66%
L11	L17	4.77	2.85	3.00	99.02%	99.02%

Table S7: **Consecutive-layer highway fractions $\mathbf{L}_i \to \mathbf{L}_{i+1}$ (Geneformer, fixed-τ).** Same protocol as Table S6 (100K gene positions per layer, fixed τ=3.0). Because the permutation null's 99th percentile PMI is below 3.0 for every long-range pair tested in Table S6, the permutation null was computed for the three long-range pairs. The curve is near-flat in the 94.7–99.1% range with no sharp drop at any single transition, supporting continuous cross-layer information flow rather than concentrated bottlenecks. The slight dip at the final layer pair (L16→L17) is consistent with the terminal-layer specialization reported in the section on layer-wise annotation in the main text.

$L_i \to L_{i+1}$	Highway%
L0 → L1	98.68%
L1 → L2	98.74%
L2 → L3	98.48%
L3 → L4	98.35%
L4 → L5	97.70%
L5 → L6	97.09%
L6 → L7	97.55%
L7 → L8	96.96%
L8 → L9	97.37%
L9 → L10	98.24%
L10 → L11	99.13%
L11 → L12	99.05%
L12 → L13	99.05%
L13 → L14	98.87%
L14 → L15	98.48%
L15 → L16	96.72%
L16 → L17	94.70%

Table S8: **scGPT cross-layer information highways.** Same methodology as Table S5. Note the progressive drop in downstream connectivity.

Layer Pair	PMI Edges	Upstream	Downstream	Max PMI
L0 → L4	75,305	1,935/2,027 (95.5%)	1,960/2,048 (95.7%)	9.15
L4 → L8	61,263	1,955/2,048 (95.5%)	1,723/2,048 (84.1%)	9.26
L8 → L11	45,258	1,773/2,048 (86.6%)	1,289/2,048 (62.9%)	10.78

Table S9: **Top cross-layer biological cascades.** Strongest annotated PMI connections between layer pairs. "Unlabeled" = the target feature lacks direct ontology annotation.

Pair	Source Feature	Target Feature	Biological Logic	PMI
L0→L5	Protein Processing in ER	*unlabeled*	ER stress cascade	11.10
L0→L5	mTORC1 Regulation	Autophagy	mTORC1→autophagy	9.55
L0→L5	Wnt Signaling	*unlabeled*	Wnt pathway processing	9.48
L5→L11	Protein Polyubiq.	*unlabeled*	Protein quality control	10.87
L5→L11	Translation	*unlabeled*	Translational regulation	10.35
L5→L11	RNA Splicing (neg. reg.)	*unlabeled*	Post-transcriptional	10.21
L11→L17	Protein Modification	Angiogenesis (pos. reg.)	PTM→phenotype	10.62
L11→L17	COPII Vesicle Budding	Thermogenesis	Secretory→metabolic	10.29
L11→L17	Actomyosin Org.	Cell Locomotion (neg. reg.)	Structure→motility	10.14

Table S10: **PMI co-activation graph dimensions and peak memory (Geneformer layers 0 and 11).** Recomputed on a deterministic 500,000-position subsample (seed 42) of the 4,056,351 K562 token positions at each layer, with `MIN_COACTIVATION`=20 and `PMI_THRESHOLD`=2.0 to match the main-text co-activation pipeline (main-text numbers use the full 4M positions with `MIN_COACTIVATION`=50). N_{feat} = alive features at the layer. Peak RSS measured with `psutil.Process().memory_info().rss`. The dense ($N \times N$) coactivation matrix accounts for ≈85 MB of the budget; the remaining ∼700 MB is the subsampled TopK index buffer and PyTorch/Python overhead.

Layer	N_{feat}	Subsampled positions	$\|E\|$ (after thresh.)	Coact matrix (MB)	Peak RSS (MB)	Encode (s)	PMI (s
0	4,608	500,000	435,252	85	811	100.1	11.
11	4,608	500,000	356,894	85	864	93.6	10.

Table S11: **Leiden resolution sweep at Geneformer layers 0 and 11.** Module count ($\geq$3 features), mean module size, coverage (fraction of graph nodes in $\geq$3-feature modules), and Adjusted Rand Index vs. the baseline γ=1.0 partition for $\gamma \in \{0.5, 0.75, 1.0, 1.5, 2.0\}$. **Note on comparison with Table S4:** this sweep is run on a deterministic 500K-position subsample with `MIN_COACTIVATION`=20 (see caption of Table S10), whereas the main-text module counts in Table S4 use the full 4,056,351 positions with `MIN_COACTIVATION`=50. The different PMI significance thresholds produce slightly different partition sizes; at γ=1.0 the sweep gives 8 modules at L0 and 9 at L11, versus 6 and 8 in Table S4. The partition is *not* stable in a strong ARI sense (ARI between γ=1.0 and its immediate neighbors is in the 0.17–0.26 range); instead, the co-activation graph exhibits a nested hierarchy in which low γ collapses the graph into $\sim$2 very large communities and high γ fragments each main community into finer subcommunities. The qualitative biological identities of modules (cell cycle, translation, metabolism, etc.) are preserved across the range; our main-text module counts correspond to the resolution-1.0 cut of this hierarchy.

Layer	γ	Modules ($\geq$3)	Mean size	Coverage	ARI vs γ=1.0
0	0.50	2	2,287.5	100.0%	+0.027
0	0.75	3	1,525.0	100.0%	+0.259
0	1.00	8	571.9	100.0%	+1.000
0	1.50	25	183.0	100.0%	+0.252
0	2.00	48	95.3	100.0%	+0.117
11	0.50	2	2,282.0	100.0%	+0.013
11	0.75	3	1,521.3	100.0%	+0.180
11	1.00	9	507.1	100.0%	+1.000
11	1.50	31	147.2	100.0%	+0.171
11	2.00	56	81.5	100.0%	+0.085

Table S12: **SAE hyperparameter ablation at Geneformer layer 11.** Grid over expansion ratio $\in \{2\times, 4\times, 8\times\}$ and TopK sparsity $k \in \{16, 32, 64\}$, trained on the same 1M-position K562 subsample used in the main atlas (Methods), evaluated on 100K held-out positions. Columns: expansion, k, total features, variance explained on held-out positions, number of dead features (no activation on the 100K hold-out), mean absolute decoder–decoder cosine similarity on a 5,000-pair sample (a proxy for quasi-orthogonality), and Leiden module count ($\geq$3 features, γ=1.0) on a 200K-position co-activation graph. The baseline configuration ($4\times$, k=32) sits at the center of the grid and does not produce an unusual point: variance explained increases monotonically with both expansion and k, module count is within 6–12 across the entire grid, and quasi-orthogonality is preserved everywhere ($\langle|\cos|\rangle \lesssim 0.05$). The main-text qualitative conclusions—substantial superposition, organized biological structure, modular co-activation—are preserved across the grid. We report the structural metrics here; full downstream annotation rate and TF specificity per grid cell are not reported here (see the Methods of the main text).

Expansion	k	N_{feat}	VarExpl	Dead	$\langle\|\cos\|\rangle$	Modules ($\geq$3)
$2\times$	16	2,304	0.706	95	0.0421	9
$2\times$	32	2,304	0.752	11	0.0383	10
$2\times$	64	2,304	0.795	1	0.0338	8
$4\times$	16	4,608	0.790	228	0.0394	12
$4\times$	**32**	**4,608**	**0.819**	**45**	**0.0365**	**7**
$4\times$	64	4,608	0.846	1	0.0327	7
$8\times$	16	9,216	0.846	559	0.0360	12
$8\times$	32	9,216	0.866	112	0.0345	8
$8\times$	64	9,216	0.884	8	0.0329	6

Bold row: main-atlas baseline used in Table 1 of the main text and throughout the paper.

Table S13: **Sensitivity of the SAE-vs-SVD comparison to the cosine threshold (Geneformer, all 18 layers).** For each layer and each cosine threshold $\tau \in \{0.3, 0.5, 0.7, 0.9\}$: percentage of alive SAE features whose decoder column has maximum absolute cosine above τ with any of the top-50 SVD axes (columns "aligned%"), and percentage of ontology enrichments found exclusively in the non-aligned ("novel") set (columns "%enrich novel"). The value τ=0.5 reproduces the per-layer SVD-aligned counts in Table 1 of the main text (the pipeline actually used τ=0.5; the pipeline threshold is τ=0.5). Across the full range, $> 98\%$ of ontology enrichments live exclusively in the non-aligned set, at every threshold tested.

	Aligned %				% enrich. from non-aligned			
Layer	τ=0.3	τ=0.5	τ=0.7	τ=0.9	τ=0.3	τ=0.5	τ=0.7	τ=0.9
0	4.19	0.89	0.02	0.00	98.88	99.98	100.00	100.00
1	3.45	0.59	0.07	0.00	99.54	99.98	100.00	100.00
2	3.73	0.63	0.04	0.00	99.49	99.93	100.00	100.00
3	3.26	0.26	0.00	0.00	99.27	100.00	100.00	100.00
4	2.69	0.28	0.00	0.00	99.48	99.97	100.00	100.00
5	2.63	0.17	0.00	0.00	98.92	100.00	100.00	100.00
6	2.39	0.11	0.00	0.00	98.62	99.99	100.00	100.00
7	2.08	0.13	0.00	0.00	98.30	99.99	100.00	100.00
8	1.48	0.07	0.00	0.00	99.40	99.98	100.00	100.00
9	1.09	0.09	0.00	0.00	99.36	99.99	100.00	100.00
10	1.09	0.15	0.00	0.00	99.63	100.00	100.00	100.00
11	0.74	0.09	0.00	0.00	99.28	100.00	100.00	100.00
12	0.65	0.07	0.00	0.00	99.26	100.00	100.00	100.00
13	0.95	0.07	0.00	0.00	98.75	100.00	100.00	100.00
14	0.93	0.04	0.00	0.00	97.90	99.55	100.00	100.00
15	1.09	0.11	0.00	0.00	98.28	99.03	100.00	100.00
16	1.00	0.11	0.02	0.00	98.73	99.02	99.47	100.00
17	1.63	0.26	0.02	0.00	98.68	99.99	100.00	100.00

Table S14: **Capacity-matched comparison against the principal subspace.**

Layer	SAE VE	k^*	k_{50}	k_{90}	ρ_{100}	align. top-50 (%)	align. full (%)
0	0.848	403	247	764	0.207	0.98	0.98
1	0.854	388	233	747	0.221	0.65	0.65
2	0.857	362	218	725	0.231	0.67	0.67
3	0.861	349	211	711	0.241	0.33	0.33
4	0.862	334	199	692	0.253	0.41	0.41
5	0.856	317	190	671	0.260	0.13	0.13
6	0.842	308	191	680	0.257	0.04	0.04
7	0.824	302	191	693	0.265	0.11	0.11
8	0.814	291	185	684	0.284	0.07	0.07
9	0.811	295	181	707	0.309	0.09	0.09
10	0.818	305	181	727	0.318	0.15	0.15
11	0.823	308	178	734	0.332	0.09	0.09
12	0.818	289	171	722	0.351	0.09	0.09
13	0.809	269	159	688	0.373	0.07	0.07
14	0.808	261	155	674	0.376	0.04	0.04
15	0.795	265	167	692	0.349	0.13	0.13
16	0.775	252	165	683	0.346	0.11	0.11
17	0.775	250	152	664	0.372	0.26	0.26

k^* is the truncated-SVD rank whose cumulative eigenvalue mass equals the SAE's variance explained. k_{50} and k_{90} are the subspace sizes at which the median decoder direction reaches 50% and 90% of its norm; for a random direction these are 576 and 1,037. The last two columns give the percentage of features aligned above $|\cos| = 0.5$ with any of the leading 50 axes and with the complete basis.

Table S15: **Dictionary atoms and distinct gene-level programs.**

Layer	Atoms	Distinct programs	Atoms/program	Mean $\lvert\cos\rvert$	Welch bound
0	4,608	1,301	3.54	0.0321	0.0255
1	4,606	1,290	3.57	0.0334	0.0255
2	4,601	1,276	3.61	0.0344	0.0255
3	4,595	1,114	4.12	0.0343	0.0255
4	4,583	999	4.59	0.0344	0.0255
5	4,576	814	5.62	0.0351	0.0255
6	4,580	774	5.92	0.0351	0.0255
7	4,591	811	5.66	0.0353	0.0255
8	4,586	817	5.61	0.0354	0.0255
9	4,595	975	4.71	0.0359	0.0255
10	4,602	1,241	3.71	0.0360	0.0255
11	4,598	1,252	3.67	0.0361	0.0255
12	4,592	1,332	3.45	0.0370	0.0255
13	4,583	1,179	3.89	0.0384	0.0255
14	4,568	1,133	4.03	0.0390	0.0255
15	4,543	1,199	3.79	0.0366	0.0255
16	4,538	2,048	2.22	0.0366	0.0255
17	4,580	1,360	3.37	0.0379	0.0255
All	**82,525**	**18,711**	**4.41**		

Programs are communities of the top-20 gene-set similarity graph at Leiden resolution 1.0; the total is 18,711 and varies by less than 0.3% over resolutions 0.5–2.0. The Welch bound is the smallest achievable maximum coherence for that many unit vectors in d dimensions.

Table S16: **Run-to-run stability of the dictionaries.**

Layer	Runs	Variance explained	Dead features	Mean best $\lvert\cos\rvert$	$\lvert\cos\rvert > 0.9$ (%)	Modules
0	6	0.8454 ± 0.0003	2 ± 1	0.282	0.8	7.2 ± 0.4
11	6	0.8200 ± 0.0003	32 ± 4	0.294	0.6	8.3 ± 1.0

Runs differ in weight initialisation and batch order with the training subsample held fixed, plus one run per layer with a different subsample. Dictionary agreement is the mean over atoms of the best decoder cosine to any atom of the other run.

Table S17: **Cross-model comparison on matched inputs at matched relative depth.**

Rel. depth	GF / scGPT layer	Geneformer/TS	scGPT/TS	scGPT/TS matched	Geneformer/K562
0.00	L0 / L0	0.7731	0.7258	0.7676	0.8444
0.29	L5 / L3	0.7749	0.7729	0.7827	0.8528
0.65	L11 / L7	0.7596	0.8515	0.8171	0.8187
1.00	L17 / L11	0.7241	0.8862	0.8552	0.7713

Variance explained on held-out positions. The two models are compared on the same Tabula Sapiens cells at matched relative depth (layer index divided by depth minus one). The matched-protocol column retrains the scGPT dictionaries under the same 1M-position subsample used for Geneformer. The final column gives the Geneformer dictionaries on their own training distribution, which bounds how much of any cross-model difference is attributable to input distribution.

Table S18: **How many perturbed transcription factors are evaluable.**

Reference	TFs perturbed	≥1	≥3	≥5	≥10	≥20	Median
TRRUST	73	55	21	9	3	1	1.0
DoRothEA A+B+C	21	17	14	13	13	10	15.0
DoRothEA (all)	30	27	22	18	18	17	146.5

Of the perturbation targets with at least 50 cells, the table counts how many are transcription factors in each reference network and, of those, how many have at least n of their curated targets among the genes the assay measures. A factor whose curated targets are not measured cannot show target-specific behaviour under any method.

Table S19: **Target recovery under one evaluation framework.** Transcription factors with significant target enrichment (FDR < 0.05).

Method	TRRUST	DoRothEA A+B+C	DoRothEA (all)
Differential expression of the knockdown	1/9 (11.1%)	3/13 (23.1%)	8/18 (44.4%)
SAE feature response	0/9 (0.0%)	0/13 (0.0%)	0/18 (0.0%)
GENIE3	1/9 (11.1%)	1/13 (7.7%)	0/18 (0.0%)
GRNBoost2	1/9 (11.1%)	1/13 (7.7%)	1/18 (5.6%)
Co-expression (Pearson)	2/9 (22.2%)	2/13 (15.4%)	2/18 (11.1%)
Co-expression (Spearman)	2/9 (22.2%)	2/13 (15.4%)	1/18 (5.6%)
Geneformer gene embeddings	1/9 (11.1%)	1/13 (7.7%)	0/18 (0.0%)
Random gene sets	0/9 (0.0%)	0/13 (0.0%)	0/18 (0.0%)

Each method emits a ranked list of 100 putative targets per transcription factor; enrichment for that factor's curated targets is tested by hypergeometric test with Benjamini–Hochberg correction across the panel, with the universe set to the genes the assay measures. Differential expression of the knockdown is perturbation-aware and bounds what these data support.

Table S20: **Unit of replication in the perturbation analysis.** Transcription factors with the largest cell-level response.

Target	Cells	Responding (cell)	Responding (position)	Inflation	ICC	Self detected
GATA1	95	30	28	0.9	0.000	no
MAX	161	7	4	0.6	0.000	yes
MYBL2	81	1	0	0.0	0.000	no
TBP	143	0	0	0.0	0.000	no
SRF	120	0	0	0.0	0.000	no
THAP1	107	0	0	0.0	0.000	no
TFDP1	140	0	0	0.0	0.000	no
CEBPZ	104	0	0	0.0	0.000	no
STAT5A	170	0	0	0.0	0.000	no
E2F6	140	0	0	0.0	0.000	no
THAP11	60	0	0	0.0	0.000	no
TERF1	72	0	0	0.0	0.000	no
BRF2	170	0	0	0.0	0.000	no
ATF4	59	0	0	0.0	0.000	no
SP2	115	0	0	0.0	0.000	no
SETDB1	123	0	0	0.0	0.000	no
HINFP	157	0	0	0.0	0.000	no
BRCA1	87	0	0	0.0	0.000	no
DNMT1	161	0	0	0.0	0.000	no
ERCC2	116	0	0	0.0	0.000	no

Responding features under a cell-level Mann–Whitney test against a position-pooled test on exactly the same cells. The intraclass correlation is the between-cell share of feature-activation variance (median across factors 0.000). Median inflation of the responding-feature count under position pooling: 0.0×. The last column records whether the knocked-down gene itself appears in the predicted set, a positive control for detection power.

Table S21: **Detection of the perturbed gene by each readout, on a common evaluable set.**

Cell line	Evaluable	DE detects	SAE detects	DE top-100	SAE top-100	Median DE rank
K562	16/20	15/16	1/16	16/16	1/16	1
RPE1	16/16	13/16	0/16	12/16	0/16	10
Jurkat	14/16	14/14	1/14	14/14	1/14	1
HepG2	15/16	13/15	0/15	14/15	0/15	1
Pooled	**61/68**	**55/61**	**2/61**	56/61	2/61	2

A target is evaluable only if the perturbed gene is measured in the assay and present in the model gene vocabulary, so that either readout could recover it; excluded targets are dropped from both numerator and denominator of both readouts. Differential expression detects the gene when its own expression differs between its perturbed cells and the non-targeting cells of the same line (Mann–Whitney over cells, Benjamini–Hochberg across all measured genes, FDR < 0.05). The feature readout detects it when the gene appears among the top-20 activating genes of at least one responding feature. The matched columns score both by the same rule: the perturbed gene falls within the first 100 entries of each readout's ranked gene list. The final column is the median rank of the perturbed gene among all measured genes by effect size.

Table S22: **Sample-size sweep.** Perturbation response as a function of cells per target.

Cells per target	Responding features	Detects knockdown (%)	Nominal enrichment (%)	Draws
10	0.56	5.0	7.7	280
20	0.65	6.8	15.5	280
50	0.65	7.1	27.5	280
100	0.61	7.1	26.9	280

Restricted to the 14 transcription factors with at least 100 cells, so sample size is not confounded with which factors are available. Twenty bootstrap draws per factor per sample size. Enrichment p-values are nominal; under Benjamini–Hochberg across the panel no sample size yields a significant factor.

Table S23: **Regulatory specificity across four cell lines and two dictionaries.**

Cell line	Dictionary	TFs tested	Evaluable	Specific	Median responding
K562	multi-line control	20	18	0/18 (0.0%)	0
K562	+ primary tissue	20	18	0/18 (0.0%)	0
RPE1	multi-line control	16	16	0/16 (0.0%)	0
RPE1	+ primary tissue	16	16	0/16 (0.0%)	0
Jurkat	multi-line control	16	16	1/16 (6.2%)	0
Jurkat	+ primary tissue	16	16	0/16 (0.0%)	0
HepG2	multi-line control	16	16	0/16 (0.0%)	0
HepG2	+ primary tissue	16	16	0/16 (0.0%)	0

Perturbed and non-targeting cells are drawn from the same line in every row, so a difference between them cannot reflect cell-line composition. Specificity is the fraction of evaluable transcription factors whose predicted target set is enriched for its curated targets (DoRothEA, all confidence levels; FDR < 0.05).

Table S24: **Causal ablation under three feature-selection rules and three evaluation gene sets.**

Selection arm	Features	top-20 $\cap$ term	Held-out term	Matched random	p
richly annotated	40	37.3	1.04	0.89	0.006
random annotated	40	56.9	0.94	0.91	0.765
random feature	26	66.9	0.97	0.89	0.340

Median specificity ratio, mean $|\Delta\text{logit}|$ on the gene set over the same quantity on all other positions, at layer 11 in k562 cells with 60 cells per feature. The held-out set is the annotated term's genes excluding the feature's own top-20; the matched random set is drawn per feature at the same size and mean-activation decile. The last column is a Wilcoxon signed-rank test of held-out against matched random. 14 of 119 features admit no held-out set because their annotated term lies entirely within their top-20 genes.

Table S25: **Cell-type and perturbation-response tests for unannotated features (Geneformer layers 0, 5, 11, 17).** Columns: #unann = number of features with zero ontology enrichments; CT% unann = fraction of those that are enriched for ≥1 of the 56 Tabula Sapiens cell types (Fisher's exact, BH FDR<0.05); CT% ann = same for the annotated population; CT% null = expected fraction for a matched random subset of size #unann drawn from all alive features; Pert% unann = fraction of unannotated features that appear in the top-20 responding list for ≥1 of the 100 Replogle CRISPRi targets with |effect| > 0.5; Pert% ann and Pert% null defined analogously. Cell-type rates are uniformly high (≈ 90–95%) for both populations, reflecting that nearly every SAE feature at these layers is cell-type-selective in Tabula Sapiens; this test does not separate the two populations cleanly. The perturbation test does separate them: at layer 11 unannotated features are ∼15× more likely than annotated features to appear in a top-20 responding list, providing non-tautological evidence that a subset of unannotated features carry real perturbation-response signal.

Layer	#unann	CT% unann	CT% ann	CT% null	Pert% unann	Pert% ann	Pert% null
0	1,906	94.96%	94.52%	94.70%	0.42%	0.22%	0.30%
5	2,225	91.73%	95.80%	93.85%	2.20%	1.72%	1.93%
11	2,025	92.94%	95.74%	94.53%	2.32%	0.15%	1.11%
17	2,454	89.41%	94.89%	91.96%	3.63%	3.81%	3.70%

Table S26: **Unannotated feature analysis (gene-set clustering).** Co-activate = unannotated feature belongs to a co-activation module containing annotated features. Isolated = no module membership. A small number of features (3 at L11, 17 at L17) belong to modules containing only unannotated features and are excluded from both columns. Clusters = standalone gene-set clusters among unannotated features. The co-activate/isolated column is weak evidence (modules cover 96–99.5% of all features so co-membership is nearly unavoidable) and is not diagnostic, because modules cover 96–99.5% of all features; the tests in Table S24 are the informative ones.

Layer	Annotated	Unannotated	Clusters	Co-activate w/ Annotated	Isolated
0	2,702	1,906	15 (48 feats)	1,876 (98.4%)	30 (1.6%)
5	2,383	2,193	19 (69 feats)	2,090 (95.3%)	103 (4.7%)
11	2,583	2,015	11 (47 feats)	1,984 (98.5%)	28 (1.4%)
17	2,154	2,426	12 (58 feats)	2,334 (96.2%)	75 (3.1%)

Table S27: **scGPT feature mutual-information with tissue vs. cell type.** 150K-position subsample at each of scGPT layers {0, 4, 7, 11}. "Tissue-dominant" = feature whose per-feature MI with tissue (3-way: immune/kidney/lung) exceeds its MI with cell type (56-way). Tissue here is a proxy for sample-level batch because the cached scGPT extraction metadata records only {tissue, cell_type}, not donor ID or sequencing assay. Across all four layers, mean MI with cell type is $\sim$2.5$\times$ larger than mean MI with tissue, and essentially zero features are tissue-dominant.

Layer	Alive feats	Mean MI(tissue)	Mean MI(cell_type)	Tissue-dominant	Frac tissue-dominant
0	1,990	0.0003	0.0008	0	0.00%
4	2,038	0.0005	0.0015	0	0.00%
7	1,999	0.0009	0.0023	0	0.00%
11	1,766	0.0016	0.0040	1	0.06%

Supplementary Figures

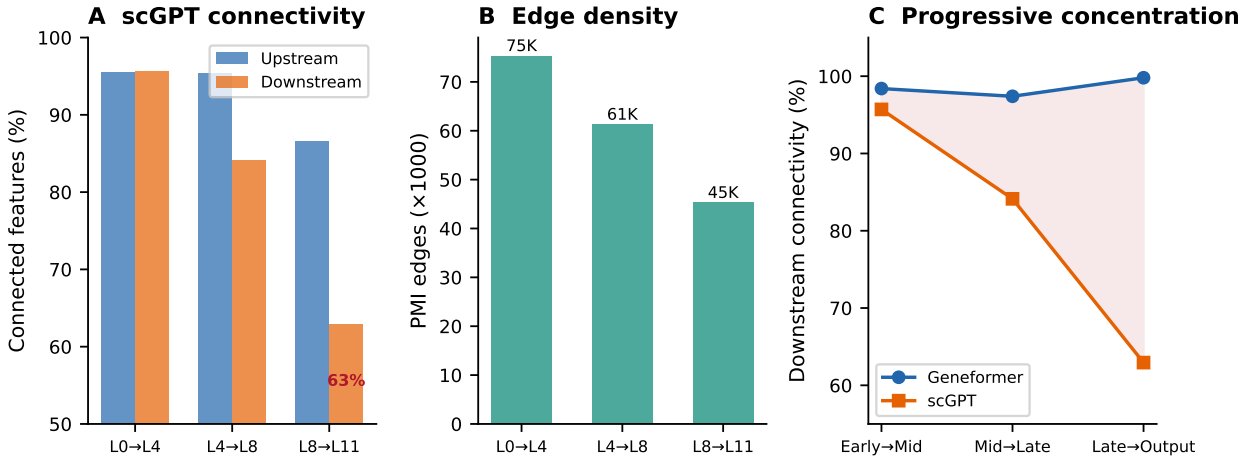
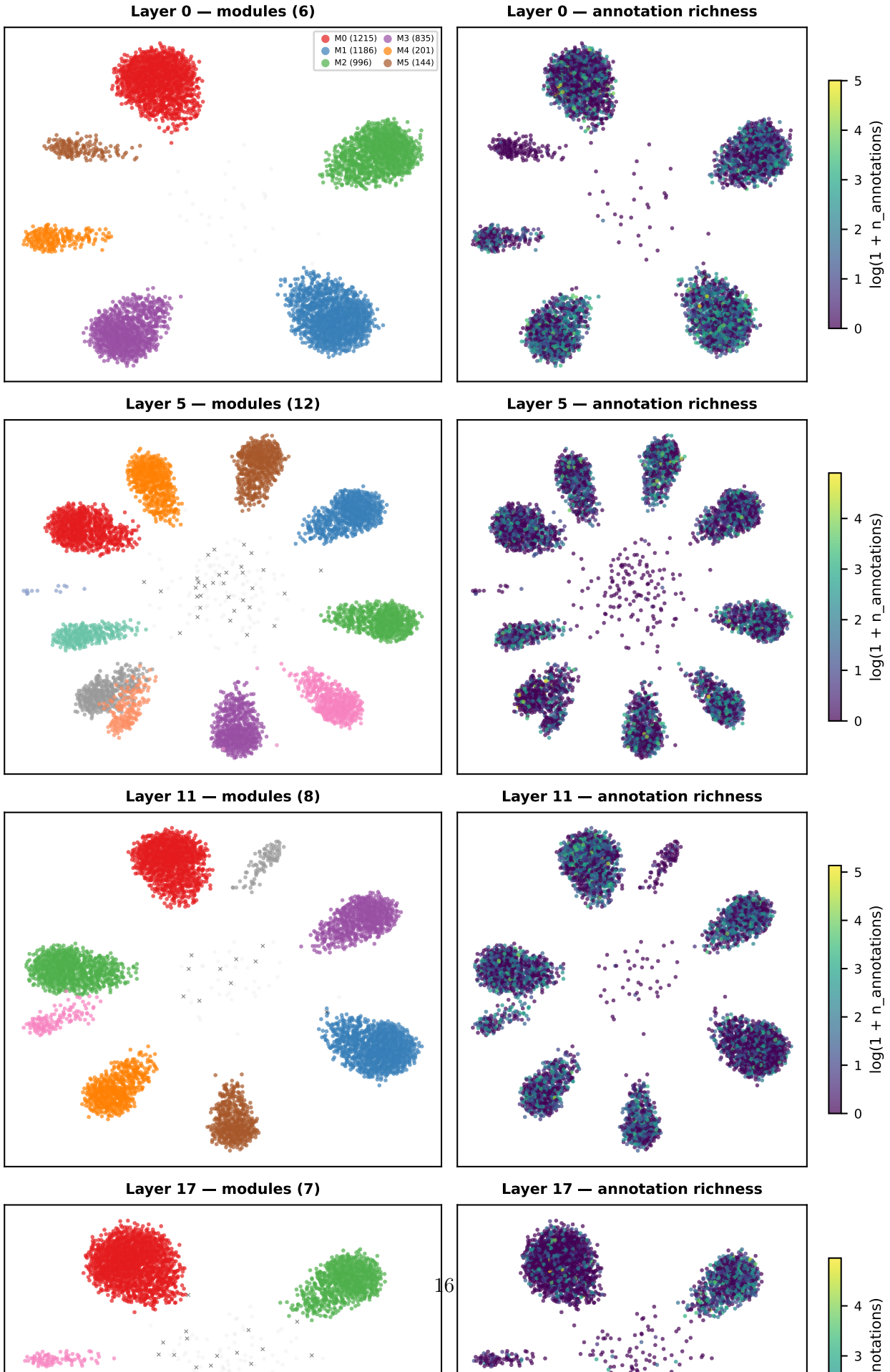


Figure S1: **scGPT cross-layer connectivity.** Upstream connectivity stays high while downstream connectivity falls with depth, a pattern Geneformer does not show.


Layer 0 — modules (6)
M0 (1215)
M1 (1186)
M2 (996)
M3 (835)
M4 (201)
M5 (144)
Layer 0 — annotation richness
log(1 + n_annotations)
Layer 5 — modules (12)
Layer 5 — annotation richness
log(1 + n_annotations)
Layer 11 — modules (8)
Layer 11 — annotation richness
log(1 + n_annotations)
Layer 17 — modules (7)
Layer 17 — annotation richness

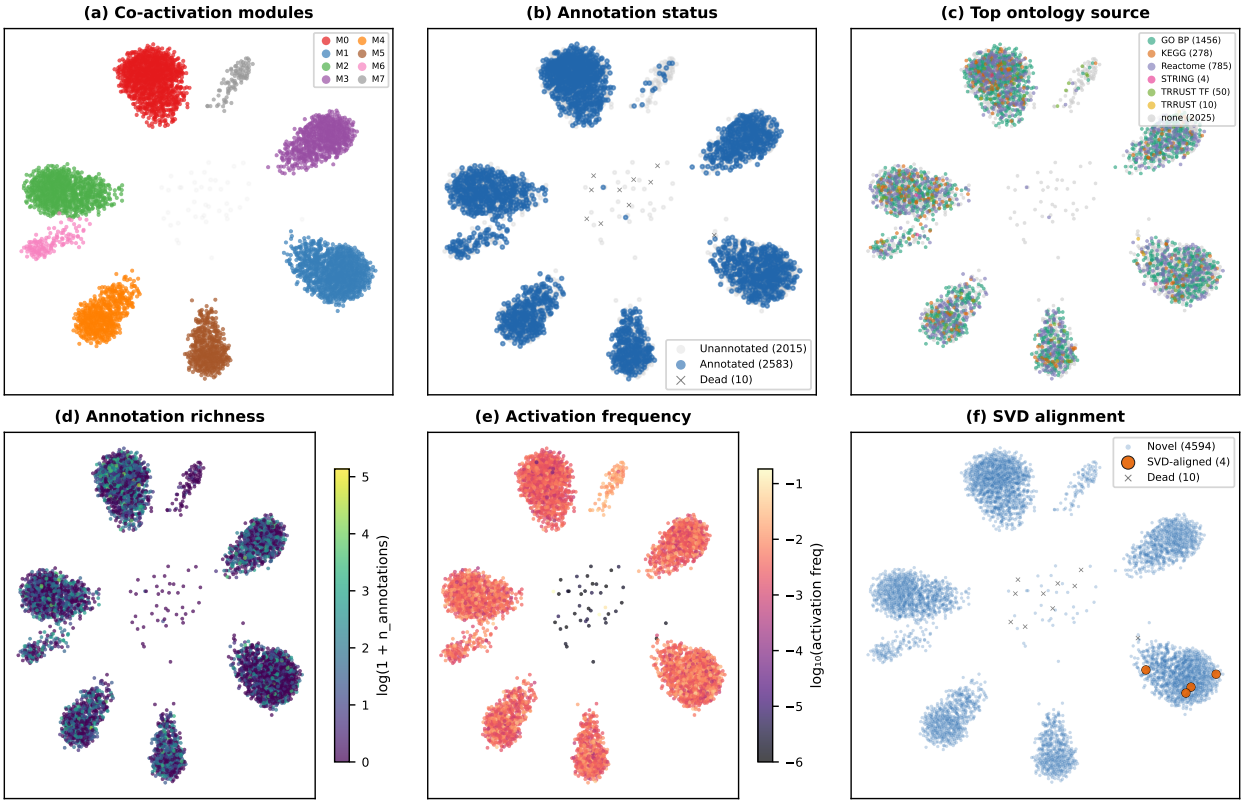


Figure S3: **Layer-11 co-activation graph in detail.** Annotated features distribute across all communities while sparsely annotated features concentrate centrally.

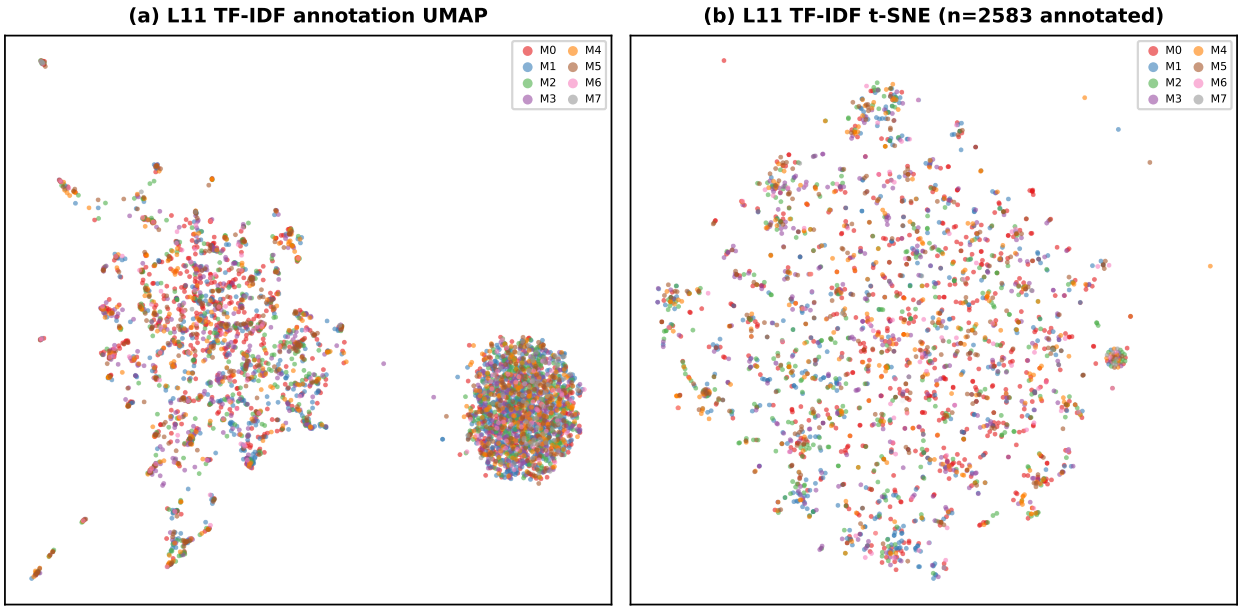


Figure S4: **Annotation-based projections.** TF-IDF weighted ontology vectors projected with UMAP and t-SNE, an independent check that module structure reflects biological similarity.